

MATHEMATICS CLASS – X WBBSE 2022

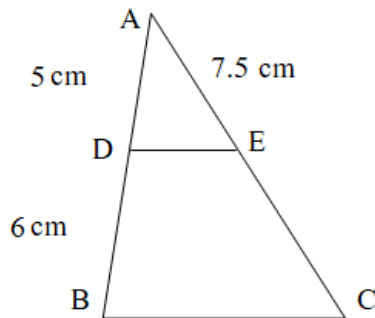
1. Choose the correct option in each case from the following questions: (1X6=6)
- At present the population of a village is P and if rate of increase of population per year $2r\%$, after n years the population will be:
(a) $P\left(1+\frac{r}{100}\right)^n$ (b) $P\left(1+\frac{r}{50}\right)^n$ (c) $P\left(1+\frac{r}{100}\right)^{2n}$ (d) $P\left(1-\frac{r}{100}\right)^n$
 - Fatima, shreya and smita started a business by investing total of Rs 6000. After a year Fatima, Shreya and Smita get profit share of Rs. 50, Rs. 100 and Rs. 150 respectively. Smita invested in this business:
(a) Rs. 1000 (b) Rs. 2000 (c) Rs. 3000 (d) Rs. 4000
 - $A:B=2:3$, $B:C=5:8$, $C:D=6:7$, then $A:D=$
(a) 2:7 (b) 7:2 (c) 5:8 (d) 5:14
 - O is the centre of a circle and PQ is a diameter. R is a point on the circle such that $PR = RQ$, then the value of $\angle RPQ$:
(a) 30° (b) 90° (c) 60° (d) 45°
 - If two circles do not intersect or touch each other, then the maximum number of common tangent/tangents is/are:
(a) 2 (b) 1 (c) 3 (d) 4
 - The volume of a solid sphere having radius $2r$ units is:
(a) $\frac{32\pi.r^3}{3}$ cubic units (b) $\frac{16\pi.r^3}{3}$ cubic units
(c) $\frac{8\pi.r^3}{3}$ cubic units (d) $\frac{64\pi.r^3}{3}$ cubic units
2. Fill in the blanks (any Five):- (1X5=5)
- The annual rate of compound interest is $r\%$ and if the first year principal is P , then 2nd year principal is _____.
 - $7\sqrt{11}$ is an _____ number.
 - If radius of a sphere is r and volume v , then $v \propto$ _____.
 - Two triangles are similar if their corresponding sides are _____.
 - If the opposite angles of a quadrilateral be supplementary, then the vertices of the quadrilateral will be _____.
 - If the length, breadth and height of a rectangular parallelepiped are equal, then the special name of this solid is _____.
3. Write True or False (any Five):- (1X5=5)
- At least 3 persons are needed in a partnership business
 - The relation between Principal and Amount is $\text{Principal} < \text{Amount}$.
 - The two roots of the equation $x^2 = 100$ are ± 10 .
 - If a and b in inverse variation, then $\frac{a}{b} = \text{constant}$.
 - Two concentric circles have only one common tangent.
 - The height, radius and slant height of a right circular cone are always the three sides of a right angled triangle.

MATHEMATICS CLASS – X WBBSE 2022

4. Answer the following questions (any ten):-

(2X10=20)

- i. The annual interest is $\frac{1}{16}$ part of its principal, then determine the interest of Rs. 690 of 8 months.
- ii. The present population is 13310. If the population be 17280 after 3 years, what will be the rate of increase?
- iii. The ratio of capitals of A, B, C is $\frac{1}{x}, \frac{1}{y}, \frac{1}{z}$, after a year there was a loss of Rs. z, calculate the loss of C.
- iv. Rationalise the surds of denominator: $\frac{12}{\sqrt{15}-3}$
- v. The radius of the circle with the centre O is 13 cm and a chord AB with the length of 10 cm on it. Calculate the distance of the chord AB from the centre of the circle.
- vi. AOB is a diameter of a circle whose centre is O. the point lies on the circle. If $\angle OBC = 60^\circ$. Find the value of $\angle OCA$.
- vii. A circle with centre O. A point P is 26 cm away from the centre of the circle and the length of the tangent drawn from the point P to the circle is 10 cm. Calculate the length of radius of the circle.
- viii. $DE \parallel BC$ of $\triangle ABC$ where D and E are two points on AB and AC respectively. If $AD = 5$ cm, $DB = 6$ cm, and $AE = 7.5$ cm. Calculate the length of AC.



- ix. If the height of two right circular cylinders are in the ratio 1:2 and perimeter of the base are in the ratio 3:4. Find the ratio of their volumes.
 - x. If the length of radius of a sphere is increased by 50%, find how much percent will be increased of its curved surface area.
 - xi. The length of diagonal of a cube is $4\sqrt{3}$ cm. Calculate the total surface area of the cube.
5. Answer any two question:- (2X5=10)
- i. At the same rate of simple interest in percent per annum, if a principal becomes the amount of Rs. 7100 in 7 years and of Rs. 6200 in 4 years, determine the principal and the rate of interest in percent per annum.
 - ii. Three friends have started a business by investing Rs 8000, Rs. 10000 and Rs. 12000 respectively. They also took an amount as bank loan. At the end of one year they made a profit of Rs. 13400. After paying the annual bank installment of Rs. 5000 they divided the remaining money of the profit

MATHEMATICS CLASS – X WBBSE 2022

among themselves in the ratio of their capitals. Calculate the profit share of each.

- iii. Calculate the difference between compound interest and simple interest on Rs. 20000 for 2 years at 5% per annum.

6. Solve any two question:- (2X3=6)

- i. Solve $\frac{1}{a+b+x} = \frac{1}{a} + \frac{1}{b} + \frac{1}{x}$, $x \neq 0, -(a+b)$
- ii. Form the quadratic equation whose roots are -4 and 3.
- iii. If $m + \frac{1}{m} = \sqrt{3}$, then find the value of (a) $m^2 + \frac{1}{m^2}$ and (b) $m^3 + \frac{1}{m^3}$

7. Answer any two question:- (2X3=6)

- i. Find the simple value of the: $\frac{\sqrt{5}}{\sqrt{3}+\sqrt{2}} - \frac{3\sqrt{3}}{\sqrt{2}+\sqrt{5}} + \frac{2\sqrt{2}}{\sqrt{3}+\sqrt{5}}$.
- ii. If $a = \frac{\sqrt{5}+1}{\sqrt{5}-1}$ and $ab=1$, calculate $\left(\frac{a}{b} + \frac{b}{a}\right)$
- iii. If 15 farmers can cultivate 18 bighas of land in 5 days, determine by using theory of variation the number of days required by 10 farmers to cultivate 12 bighas of land.

No. of Farmers = A	Amount of Land = B Bigha	No. of days = C
15	18	5
10	12	?

8. Answer any one question:- (1X3=3)

- i. If $a:b=b:c$, prove that $\frac{abc(a+b+c)^3}{(ab+bc+ca)^3} = 1$
- ii. If $\frac{a}{1-a} + \frac{b}{1-b} + \frac{c}{1-c} = 1$, then find the value of $\frac{1}{1-a} + \frac{1}{1-b} + \frac{1}{1-c}$.

9. Answer any one question:- (1X5=5)

- i. Prove that the opposite angles of a cyclic quadrilateral are supplementary.
- ii. Prove that the perpendicular drawn to a chord which is not a diameter, from the centre of the circle, bisect the chord.

10. Answer any one question:- (1X3=3)

- i. ABCD is a cyclic quadrilateral. Chord DE is the external bisector of $\angle BDC$. Prove that AE (or produced AE) is the external bisector of $\angle BAC$.
- ii. Two chords, AB and CD of a circle with centre O, when produced intersect each other at a point P. Prove that $\angle AOC - \angle BOD = 2\angle BPC$.

11. Answer any one question:- (1X5=5)

- i. Draw a right angled triangle having two sides 4 cm and 8 cm length respectively containing right angle. Then draw the circumcircle of the right angled triangle. (Only traces of construction are required).
- ii. Draw a circle with radius 2.6 cm length and draw a tangent on this circle from an external point at a distance of 6 cm from the centre of the circle.

MATHEMATICS CLASS – X WBBSE 2022

12. Answer any four questions:-

(4X4=16)

- Half of a cuboidal water tank with length of 2.1 m and breadth 1.5 m is filled with water. If 630 litre of water is poured into the tank, then calculate the increased height of water.
- Height of a circular cylinder is twice of its radius. If the height would be 6 times of its radius, then the volume of the cylinder would be greater by 539 cubic dcm, calculate the height of the cylinder.
- In a right circular conical tent 11 persons can stay. For each person 4 sq. m. space in the base and 20 cu. m. air are necessary. Determine the height of the tent put up exact for 11 persons.
- Calculate how many spherical marbles with 1 cm radius may be formed by melting a solid sphere of iron having 8 cm of radius.
- The inner length, breadth and height of a tea box are 7.5 dcm, 6 dcm and 5.4 dcm respectively. If the weight of the box filled with tea is 52 kg 350 gm, but in empty state, its weight is 3.75 kg, then calculate the weight of 1 cubic dcm tea.

(ALTERNATIVE QUESTION FOR SIGHTLESS CANDIDATES)

11. Answer any one question:

(1X5=5)

- The lengths of sides containing right angle of a right angled triangle are given. Describe the procedure of construction of circumcircle of the triangle.
- Describe the process of drawing one tangent to a circle from an external point.

[Additional Questions for External Candidates]

13. (a) Answer any three questions:

(2X3=6)

- Find the value of a , if one root of the equation $2x^2 + ax + 8 = 0$ is 1.
- What is the rate of simple interest per annum, when the interest of some money in 10 years will be $\frac{2}{5}$ part of its amount?
- Which one of $\sqrt{8}$, $\sqrt{18}$, $\sqrt{27}$, $\sqrt{72}$ is not a similar surd?
- Find the total surface area of a cone whose diameter of the base is 20 cm and slant height is 25 cm.

(b) Answer any four questions:-

(1X4=4)

- What will be the compound interest and simple interest for one year at a fixed rate of interest on fixed sum of money?
- What is the value of semicircular angle?
- Find the third proportional of 5, 10.
- What is the name of solid which is composed of only one surface?
- What is the relation between the opposite angles of a cyclic quadrilateral?
